NH5 Highway AI

Project report

28 September 2026 · Model version v8

1. Summary

The NH5 Highway detection model **v8** is complete. It detects 12 classes - vehicles (car, motorcycle, auto, bus, truck, bicycle), people, potholes, number plates, road work, accidents and vehicle fires - and reads Indian number plates. It exceeds the accuracy target, and the conversion of the model to ONNX (FP32 and INT8) for the Rubik Pi board is complete.

Item	Result	Status
Accuracy target (mAP50 > 0.75)	**0.805** on the leak-free test set (0.818 on the full test set)	Met
Model training	100 epochs on the cleaned dataset (105k images)	Completed
ONNX conversion	FP32 ONNX and calibrated INT8 ONNX (3.7 MB, mAP50 0.805)	Completed
Website	Runs v8 with accident and fire alerts, vehicle counts per type, Indian plates	Completed
Road construction	Data added, but false detections - to be trained separately	Planned

2. Objective and scope

- **Goal:** automatic monitoring of the NH5 highway from cameras - count and classify traffic, read number plates, and raise alerts for accidents, vehicle fires, road work and potholes.
- **Target platform:** the Rubik Pi edge board (model converted to INT8 ONNX), and the project website for uploaded videos and a live camera.
- **Output of the system:** annotated video, alerts with snapshots, vehicle counts per type, number plates, and an Excel report per video.

3. How the system works

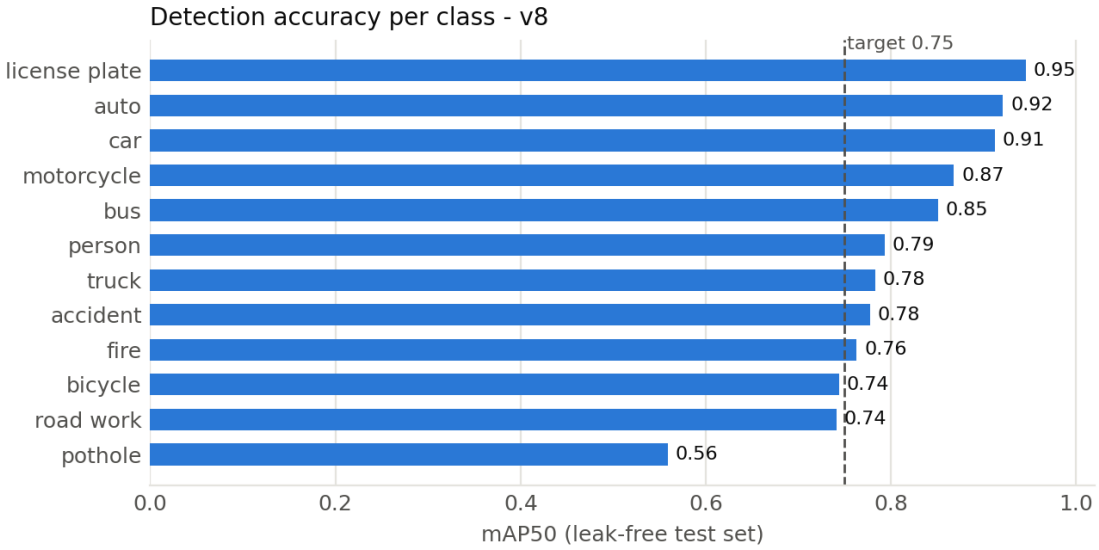
Step	What happens
1. Detection	The v8 model finds all objects of the 12 classes in each video frame.
2. Tracking	Each object keeps the same ID across frames, so a vehicle is counted once.
3. Alerts	Accident and fire must be seen steadily for about a second before an alert is raised, which avoids false alarms from a single frame.
4. Number plates	Each plate is cleaned (sharpened, straightened), read, and accepted only if it is a valid Indian plate seen the same way in several frames.
5. Report	Annotated video, alert snapshots, counts and plates are recorded for every video.

4. What the model detects

Class	Meaning	Used for
car, motorcycle, auto, bus, truck, bicycle	Vehicles (rider counted with the vehicle)	Traffic counts per type
person	Pedestrians	People count, pedestrian events
license_plate	Number plate of a vehicle	Indian plate reading
pothole	Hole / broken patch in the road surface	Road maintenance
road_construction	Road work zone (cones, barriers, workers)	Work-zone alert
accident_scene	Collision or crashed / overturned vehicle	Accident alert
fire_accident	Vehicle on fire (flames and smoke)	Fire alert

5. Accuracy

mAP50 per class (1.0 = perfect). The leak-free test uses only test images with no near-copy in the training data, so it is the most honest measure. INT8 is the compressed version converted for the board.

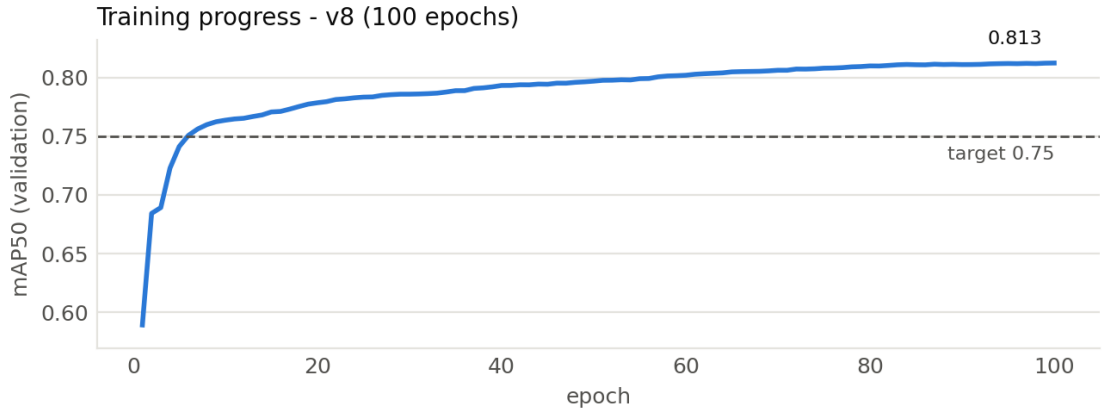


Class	Full test	Leak-free test	INT8 (board)
license_plate	0.939	0.946	0.935
auto	0.921	0.921	0.915
car	0.916	0.912	0.911
motorcycle	0.871	0.868	0.864
bus	0.851	0.851	0.850
fire_accident	0.837	0.763	0.804
accident_scene	0.802	0.777	0.772
person	0.801	0.793	0.794
truck	0.792	0.783	0.785
road_construction	0.745	0.741	0.716
bicycle	0.740	0.744	0.721
pothole	0.597	0.559	0.590
All classes	**0.818**	**0.805**	**0.805**

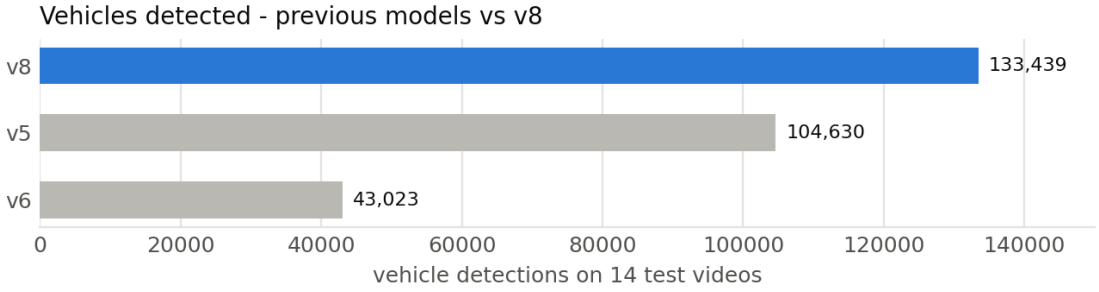
Strong: number plates, autos, cars, motorcycles, buses (0.85-0.95). Weakest: potholes (0.56-0.60), road work and bicycles (about 0.72-0.75).

6. Training

The model was trained for 100 epochs on 88,083 training images at 960 px. Accuracy passed the 0.75 target within the first 10 epochs and kept improving steadily to 0.813 (validation) at epoch 100.



7. Compared with the previous models



	v5	v6	v8
Vehicles detected (14 test videos)	104,630	43,023	**133,439**
Correct number plates read	10	31	**31**
Accident detection	not possible	yes, but misses vehicles	**yes**
Fire detection	not possible	not possible	**yes**
Count stability (lower = better)	12.1%	24.0%	**8.3%**

8. Results on test videos

Scenario	v8 result
Vehicle fires (dashcam and phone videos)	Fire detected and alert raised
Real crashes (CCTV compilation)	All crashes detected and alerted
Busy city traffic, no incidents	No false accident or fire alerts; Indian plates read correctly
Close-up of a normal car	No accident alert
Foreign (non-Indian) number plates	Correctly ignored

9. Work completed

Area	Work done
Dataset	Final dataset of 105k images; missing vehicle labels added; fire and road-work labelling rules applied
Training	YOLOv8n trained for 100 epochs at 960 px; verified on test images and 14 test videos
ONNX conversion	FP32 ONNX export, then calibrated INT8 quantization (3.7 MB) with accuracy validated against the original model
Number plates	Indian-only plate reading with image clean-up and multi-frame confirmation
Website	v8 model, accident and fire alerts, vehicle counts per type, processing speed display, plate counts

10. Milestones

Date	Milestone
Aug - Sep 2026	Earlier models (v5, v6) and first website version
24 Sep 2026	Final dataset cleaned (missing vehicle labels added, fire / road-work rules applied)
26 Sep 2026	v8 training completed (100 epochs) and verified
28 Sep 2026	ONNX conversion (FP32 + INT8) completed; website moved to v8 with fire and accident alerts and Indian-only plate reading

11. Known limitations

Limitation	Impact	Planned fix
Potholes	Misses about half	Label clean-up
Accidents on some CCTV clips	Marked 1-3 s before the impact	Relabelling
Road construction	False detections (bollards, lone signs)	Train separately
Plates from high/overhead cameras	Fewer plates read	More overhead footage
Photos of wrecked vehicles	Not always recognised	More training data
Smoke-only fires (no flames)	No fire alert	More training data

Note - road construction: road-construction data has been added to the dataset and the model detects work zones, but it still gives false detections (for example permanent roadside bollards and lone signs). Road construction **will be trained separately** because of these false detections.

12. Next steps

- **Road construction**: the data is added, but it gives false detections, so it will be trained separately.
- Improve the weaker classes (potholes, accidents on CCTV, aftermath photos) with corrected labels and more data.
- Footage from the real NH5 cameras (especially overhead views and night) would help most.

Note - lane change and wrong-direction detection: these will be checked once the camera is fixed in the position where the model needs to be tested, because both depend on the camera view (the lanes and the allowed direction of travel are set for that fixed position).